

# Aqif Ahmed

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## SUMMARY

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Software Engineering graduate building practical AI on top of pre-trained models — prompting, retrieval-augmented generation, and agents that call tools. Comfortable across the full stack an AI product needs: typed APIs, vector search, and clean deploys, not just the model call in the middle.

## TECHNICAL SKILLS

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**LLMs & Prompting:** pre-trained model selection, context/inference tradeoffs, system prompts, zero/few-shot & chain-of-thought, structured output, function calling

**RAG & Retrieval:** chunking, embedding models, semantic search, RAG vs. fine-tuning, LangChain, LlamaIndex

**Vector Databases:** Chroma, Pinecone, FAISS, Qdrant, pgvector

**Agents & Tools:** tool/function calling, ReAct loops, multi-agent workflows, MCP servers & clients

**Safety & Evaluation:** prompt-injection defense, bias & fairness, adversarial testing, constrained I/O, observability

**Engineering:** Python, FastAPI, Docker, Docker Compose, PostgreSQL, Git, Linux, JavaScript, React Native

## PROJECTS

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**InvisibleSearch** *Python, FastAPI, JavaScript, Docker* [invisiblesearch.aqifahmed.com](https://invisiblesearch.aqifahmed.com)

- Built a live search engine over GitHub: queries proxy through a FastAPI backend straight to the GitHub search index, the same pattern an LLM app uses to call an external API and shape its response.
- Mapped raw API responses into typed, paginated results behind a debounced, keyboard-navigable type-ahead — structured outputs, not loose JSON.
- Designed for GitHub's rate limits: a 60-second per-query cache, structured 429 responses with reset timestamps, and quota warnings surfaced in the UI.
- Containerized with Docker Compose; pytest suite runs against a mocked GitHub client.

**NetZero** *React Native, Expo, JavaScript* [netzero.aqifahmed.com](https://netzero.aqifahmed.com)

- Built and shipped a cross-platform expense-tracking app with full offline capability — every record persists in local storage, no connection required.
- Built around a central state layer and reusable components, with validation at the input boundary — the same discipline a retrieval layer or agent memory needs.

## EDUCATION

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### Bachelor of Science in Software Engineering

University of Central Punjab (UCP), Lahore

**Relevant Coursework:** Data Structures & Algorithms · System Design · Applied Machine Learning · Formal Methods in Software Engineering · Operating Systems · Computer Communication Networks · Software Construction & Design · Object-Oriented Programming · Advanced Web Programming

## ADDITIONAL

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**Languages:** English (professional working proficiency), Urdu (native)

**Location:** Based in Lahore; able to work on-site full time.